

John Leiga

johnleiga9@gmail.com • 209.822.9104 • [LinkedIn](#) • [Portfolio](#)

U.S. Citizen • Eligible to work on export-controlled programs

Education

B.S. Mechanical Engineering

2017 – 2021

University of California, Merced

Skills

Mechanical Design: Precision mechanical design, GD&T, ASME Y14.5, tolerance stackups, DFM, DFA, production drawings

Analysis and Validation: Stress analysis, FEA, environmental testing, vibration, shock, thermal, instrumentation, data acquisition, failure analysis

Manufacturing Systems: Root cause analysis, 5 Whys, DOE, capability analysis, process optimization, factory startup, equipment validation, CapEx, vendor coordination

Tools: SolidWorks, Autodesk Inventor, AutoCAD, Python, MATLAB, Minitab, Git

Experience

Nestle Purina PetCare Company

Bloomfield, MO — Ft. Dodge, IA — St. Louis, MO

Design Engineer

Feb 2025 – Present

- Own mechanical design and production support projects across multiple facilities, defining equipment requirements, tooling updates, and operating standards from factory needs
- Validate process equipment for factory startup, including installation readiness, startup checks, production ramp support, and troubleshooting
- Design extrusion dies, production tooling, and equipment components in Autodesk Inventor for high volume manufacturing

Management Development Associate - Engineering

May 2023 – Feb 2025

- Stabilized a 14 furnace perlite expansion facility by designing and validating a density based feedback control loop, tuning controls, and improving equipment performance
- Executed 4 to 5 factor DOEs in Minitab to tune furnace parameters, improve process stability, and support data driven production decisions
- Developed and deployed a zero waste automated startup sequence, saving \$200K+ annually across Missouri and Virginia facilities
- Executed \$500K+ in CapEx projects, supporting equipment specification, vendor coordination, installation, commissioning, and startup validation

Tavis Corporation

Mariposa, CA

Project Engineer

Sep 2021 – May 2023

- Applied first principles mechanical analysis to redesign and validate a differential pressure transducer architecture across pressure ranges, reducing mechanical cost 50% and total system cost 30% on a \$40M+ aerospace program
- Built a Microsoft Excel based nonlinear diaphragm model using advanced formulas for deflection under stress to predict diaphragm response and sensor behavior where standard FEA was insufficient, then validated predictions against test data
- Designed production ready components and assemblies in SolidWorks, including stress analysis, manufacturing drawings, and design documentation
- Developed qualification test procedures to validate hardware survival and operation under thermal, vibration, shock, and overpressure conditions
- Led internal and customer facing design reviews, aligning requirements, tradeoffs, test results, qualification data, and technical deliverables
- Managed \$10M+ aerospace hardware programs through qualification and manufacturing integration, owning requirements, schedules, and customer communication

Projects

Quantitative Research & Modeling System

- Built a modular Python machine learning system for equities and options, spanning data ingestion, feature generation, LightGBM modeling, backtesting, and live inference
- Engineered scalable Parquet pipelines, TOML configuration, walk forward testing tools, and Git based revision tracking for reproducible market research workflows